

Factory-to-Customer Shipping Route Efficiency Analysis

Nassau Candy Distributor — Final Analytical Report

Source: supplied Nassau Candy Distributor workbook and the provided Unified Mentor project brief. The project defines route analysis as Factory → Customer State/Region, with KPIs including shipping lead time, route volume, delay frequency and route efficiency.

1. Executive Summary

- 10,194 records representing 8,549 unique orders were analyzed across 5 factories, 59 states/provinces and 196 factory-to-state routes.
- Average calculated shipping lead time is 1,320.84 days and median is 1,274 days. Every record has a lead time above 365 days.
- This is driven by a source-data date mismatch: order dates are 2024–2025 while ship dates are 2026–2030. The analysis therefore describes the supplied data, but lead-time results should not be treated as real-world delivery performance until the dates are validated.
- Total Sales in the supplied workbook are \$141,783.63 and total Gross Profit is \$93,442.80.

2. Fastest Route Records

Route	Shipments	Avg_Lead_Time	Efficiency_Score
Secret Factory → New Mexico	2	906.00	100.00
Secret Factory → Nebraska	1	906.00	100.00
The Other Factory → Louisiana	1	907.00	99.86
The Other Factory → Connecticut	2	907.50	99.80
Secret Factory → Mississippi	1	908.00	99.73
Wicked Choccy's → Maine	2	908.00	99.73
Secret Factory → Louisiana	2	908.50	99.66
Secret Factory → Delaware	1	909.00	99.59
Secret Factory → South Carolina	1	909.00	99.59
Secret Factory → Minnesota	1	909.00	99.59

3. Slowest Route Records

Route	Shipments	Avg_Lead_Time	Efficiency_Score
Sugar Shack → New Jersey	1	1,642.00	0.00
Secret Factory → New Hampshire	1	1,641.00	0.14
Sugar Shack → Connecticut	1	1,641.00	0.14
Wicked Choccy's → West Virginia	2	1,639.00	0.41
Lot's O' Nuts → North Dakota	5	1,638.20	0.52
Sugar Shack → California	1	1,638.00	0.54
The Other Factory → Nevada	1	1,638.00	0.54
Secret Factory → Connecticut	1	1,638.00	0.54

Route	Shipments	Avg_Lead_Time	Efficiency_Score
Sugar Shack → Washington	1	1,638.00	0.54
Sugar Shack → Ohio	2	1,637.50	0.61

4. Regional Performance

Region	Shipments	Avg_Lead_Time	Median_Lead_Time	Sales
Gulf	1620	1,311.37	1274	22,247.26
Pacific	3253	1,322.19	1274	46,301.53
Atlantic	2986	1,322.75	1274	41,197.24
Interior	2335	1,323.09	1274	32,037.60

5. Ship Mode Performance

Ship Mode	Shipments	Avg_Lead_Time	Median_Lead_Time	Sales
Standard Class	6120	1,314.33	1274	85,490.35
Second Class	1979	1,323.85	1273	27,860.22
Same Day	547	1,333.44	1269	7,113.67
First Class	1548	1,338.28	1272	21,319.39

6. Geographic Bottlenecks

The slowest state-level averages in the supplied data are West Virginia, North Dakota, Saskatchewan, Manitoba and Iowa. Some have very low shipment counts, so they should not be interpreted as robust bottlenecks without a minimum-volume rule.

State/Province	Region	Shipments	Avg_Lead_Time
West Virginia	Atlantic	4	1,638.00
North Dakota	Interior	7	1,637.86
Saskatchewan	Pacific	2	1,457.00
Manitoba	Interior	12	1,455.33
Iowa	Interior	30	1,443.90
New Mexico	Pacific	37	1,441.84
Vermont	Atlantic	11	1,438.91
Prince Edward Island	Atlantic	10	1,420.30
South Dakota	Interior	12	1,395.92
Tennessee	Gulf	183	1,391.49

7. Data Quality & Validation

Check	Value	Status
Record count	10194	PASS
Missing Order Date	0	PASS
Missing Ship Date	0	PASS
Negative lead times	0	PASS
Order date range	2024-01-02 to 2025-12-31	INFO
Ship date range	2026-06-30 to 2030-06-28	WARNING
Lead time range	904 to 1642 days	WARNING
Rows with lead time > 365 days	10194	WARNING
Unmapped products	0	PASS

8. Recommendations

- Validate and correct the Ship Date field against the original operational source before using lead time for service-level decisions.
- After date correction, rerun route rankings, delay frequency and efficiency scores; the current 100% delay rate is a direct consequence of the future-shifted dates.
- For bottleneck analysis, require a minimum route/state shipment volume (for example, 20 records) before labeling a route a persistent bottleneck.
- Keep the factory mapping as a reference table because the raw workbook does not include a Factory field; factory is inferred from Product Name using the project brief.
- Use the Streamlit dashboard to filter by date range, region/state, ship mode and lead-time threshold as specified in the project requirements.

9. Deliverables Included

The accompanying Excel workbook contains cleaned/derived data, route KPIs, top/bottom route tables, region, state, ship-mode and factory KPIs, an executive summary, and a data-quality audit.